Scilight

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User Guide

IJGTP Class Documentation

Abstract: This guide describes the public interface of `ijgtp.cls`: preparing one manuscript or a collection of papers, setting author and publication metadata, using figures and tables, and compiling bibliographies. It describes the main commands available to authors, selected standard commands whose behavior the class changes, and the additional table column types. Internal commands containing `@` are implementation details and are not intended for use in manuscripts.

Keywords: LaTeX; IJGTP; document class; author instructions

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1. Files, packages, and class options

Place `ijgtp.cls` and `scilight.bst` in the working directory or in a directory searched by TEX. The title and footer require the image files `header`, `F1`, and `cc-by`; the ORCID command also requires `orcid`. Supply their PDF versions for PDFLaTeX and their EPS versions for the LATEX–dvips workflow. The sample plot files are examples, not class dependencies.

The class is based on `article`, defaults to 10-point text, and sets A4 journal geometry. Options are passed to `article`: for example, `\documentclass [12pt]{ijgtp}` selects 12-point text.

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Load optional packages explicitly. For example:

```
\documentclass[10pt]{ijgtp}
\usepackage{amsmath,amssymb,cmap}
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage[english]{babel}
% \usepackage{float} % only if its facilities are needed
% \usepackage{epstopdf} \epstopdfsetup{suffix=} % convert EPS figures to PDF
```

The class already loads `natbib`, `bibunits`, `etoolbox`, `geometry`, `xcolor`, `microtype`, `setspace`, `times`, `titlesec`, `array`, `lineno`, `graphicx`, `caption`, `fancyhdr`, `lastpage`, `catchfile`, and `hyperref`. Their commands remain available.

2. A minimal manuscript

Define the metadata before `\maketitle` . Each `\author` starts a new author record. Subsequent affiliation, email, ORCID, and correspondence commands apply to that author. Avoid verbatim material inside the captured abstract environment.

```
\documentclass{ijgtp}
\begin{document}
\title{Title of the manuscript}
\author{Wednesday Addams}
\affiliation{Department of Physics, Nevermore Academy}
\email{inmemoryofwednesday@gmail.com}
\corresponding
\begin{abstract}
An overview of the question, method, and results.
\end{abstract}
\keywords{gravitation; theoretical physics}
\maketitle
\section{Introduction}\label{sec:intro}
The manuscript starts here.
\section{Conclusions}
See Section~\ref{sec:intro}.
\end{document}
```

`template.tex` provides an extended example that can serve as a starting point for a manuscript.

3. Author-facing command reference

The following list covers the public commands and environments defined or given paper-specific behavior by the class. Arguments in examples are placeholders unless stated otherwise.

3.1. Basic information

`\title {title}` Sets the title for the next `\maketitle` . The title is retained after printing, so it can be reused if no replacement is supplied. Every manuscript is expected to have a title.

`\papertype {type}` Sets the designation above the title, such as `Article` or `Review`. The default is `Article`.

`\author {full name}` Appends an author. Repeat this command for additional authors. Author notes can be supplied with `\thanks {note}` inside the name.

`\authorname {surname}{initials}` Overrides the parts of the current author's name used in running heads and the suggested citation. It does not alter the displayed full name. For example, use `\authorname {van Beethoven}{L.}` after `\author {Ludwig van Beethoven}`. Without an override, the last space-separated name component is treated as the surname and earlier components supply initials. Use the override for compound surnames, suffixes, or other names that this simple rule cannot represent.

\affiliation {address} Adds an affiliation to the current author. Repeat it for multiple affiliations. Identical affiliation text is shared between authors.

\email {address} Sets the current author's email address. Only addresses of authors marked as corresponding are printed in the correspondence block. Literal underscores and escaped underscores are supported.

\orcid {identifier} Sets the current author's ORCID identifier, without the URL prefix. It produces an icon linking to the ORCID page. An empty argument omits the icon. The identifier is not validated by the class.

\corresponding Marks the current author as corresponding. It takes no argument. The class suppresses the distinguishing asterisk when all authors are corresponding. The correspondence block requires a nonempty email address from at least one corresponding author.

The commands `\authorname` , `\affiliation` , `\email` , `\orcid` , and `\corresponding` require a preceding `\author` ; otherwise, the class reports an error.

3.2. Abstract, keywords, and metadata

abstract environment Collects the abstract for the next title block. Use `\begin {abstract}` and `\end {abstract}` before `\maketitle` . A later abstract replaces an earlier one.

\keywords {list} Sets the keyword text; use semicolons or another desired separator.

\pacs {list} Sets the PACS classification text. It is printed after the keywords.

\arxivnumber {identifier} Sets an arXiv identifier without an `arXiv:` prefix or URL. Examples are `2507.22660`, `2603.12466v2`, and `gr-qc/0307012`. The class checks the identifier's syntax, not whether the record exists. Invalid input produces a warning and preserves any earlier valid identifier. The link appears in the suggested citation.

\maketitle Starts a paper, first finishing the preceding pages and floats.

\appendix Switches the current paper to appendix section numbering.

Keywords, PACS codes, and the arXiv identifier are optional. Empty or space-only keyword and PACS arguments omit their headings.

3.3. Publication identifiers

\ijgtppublished {year}{volume}{issue}{article} Sets journal publication metadata and enables the journal citation suggestion, publication running head, and DOI footer. For example:

```
\ijgtppublished {2026}{2}{1}{5}
```

This constructs `10.53941/ijgtp.2026.100005`. The DOI consists of the year, issue, and article number padded to at least five digits; the volume does not enter this identifier. Use actual assigned publication data. Arguments must contain decimal digits only. The year must be at least 2025, and the other values must be positive. Each value must fit TeX's integer range. Leading zeros are accepted and normalized. Invalid input produces a warning and preserves any earlier valid values.

\receiveddate {date} Sets the received-date text.

\reviseddate {date} Sets the revised-date text.

\accepteddate {date} Sets the accepted-date text.

\publisheddate {date} Sets the publication-date text.

Date text is printed as supplied: no date parsing or chronological validation is performed. Without publication metadata, the running head indicates a manuscript prepared for the journal. The copyright year is the supplied publication year, if available; otherwise, it is the current compilation year.

3.4. Bibliography commands and formatting hooks

\bibliography {database} Prints the current bibliography using the named `.bib` database, then closes that bibliography unit. Omit the filename extension; multiple database names may be comma-separated.

\bibliographystyle {style} Selects the current paper's BibTEX style. Place it after `\maketitle` and before `\bibliography` . It may follow citations.

\bibnumfmt {number} The class redefines this `natbib` formatting hook to print a bibliography number followed by a period. Authors normally do not call it.

\bibsection The class redefines this `natbib` hook to create the unnumbered references heading and its table-of-contents entry.

Use the loaded `natbib` commands, such as `\cite {key}`, `\cite {first,second}`, and `\nocite {key}`, in the usual way. The class uses numeric citations with sorting and compression. `\nocite {*}` includes all entries from the selected database.

3.5. Additional table column types

The class defines `L{width}`, `C{width}`, and `R{width}`: fixed-width, vertically centered paragraph columns with left, centered, and right horizontal alignment, respectively. For example:

```
\begin{tabular}{L{4cm}C{2cm}R{2cm}}
Quantity & Symbol & Value \\
Decay time & $\tau$ & 1.0 \\
\end{tabular}
```

The public helper `\PreserveBackslash {alignment}` applies an alignment command while retaining the meaning of the table row terminator `\\`. The three column types use it internally; ordinary tables do not need to call it directly.

4. Two-column mode and layout features

Two-column mode works similarly to that of the REVTEX class, whose page-grid routines are integrated into this class. Standard `figure` and `table` floats use one column; `figure*` and `table*` span both columns.

`\twocolumn` switches the following text to two columns on the current page. `\onecolumn` balances the preceding columns and continues at full width below them. A page break occurs naturally if there is insufficient space. Repeating the current mode does not force a page break. The optional form `\twocolumn [full-width material]` places that material above the new columns. Similarly, `\tableofcontents` keeps the current column mode. `\tableofcontents *` uses the full width and then restores the preceding column mode, without forcing a new page. Pending column floats may be deferred past a switch. A title, its contents, and the two-column body can share a page.

```
Full-width introduction.
\twocolumn
Text in two columns.
\onecolumn
Full-width text below the balanced columns.
```

The `widetext` environment temporarily spans both columns and restores two-column text afterwards. In one-column mode, `widetext` adds no rules or layout changes. Its contents can span multiple pages.

```
\begin{widetext}
Full-width text or a wide equation.
\end{widetext}
```

The loaded `geometry` package provides `\newgeometry` , `\savegeometry` , `\loadgeometry` , and `\restoregeometry` . A user-selected layout remains active until changed or until the next `\maketitle` , which restores the journal layout. Title-page and body margins differ; the normal body layout resumes after the first physical page unless the user explicitly changes geometry.

The loaded `lineno` package provides `\linenumbers` and `\nolinenumbers` . The title material is not line-numbered. If line numbering was enabled before `\maketitle` , it resumes afterwards; the running line-number sequence restarts for each paper.

5. Several papers in one document

Repeat the metadata and `\maketitle` for each paper. After a title is printed, the class clears the authors, affiliations, emails, ORCIDs, correspondence flags, abstract, keywords, PACS, dates, publication metadata, and arXiv identifier. Each paper has independent numbering and cross-references for equations, figures, tables, and sections. A bibliography can be shared by several papers or provided separately for each paper, using either the `\bibliography` command with BibTEX or the `thebibliography` environment.

5.1. Layout

Each paper starts on a new page. An optional document-wide contents page precedes the first title:

```
\begin{document}
\tableofcontents % all papers and sections
\title{First paper}\author{Alice Smith}
\begin{abstract}First abstract.\end{abstract}
\maketitle
\tableofcontents % this paper only
\section{Introduction}\label{sec:intro}
First paper text.
\bibliography{first-references}

\title{Second paper}\author{Bob Jones}
\begin{abstract}Second abstract.\end{abstract}
\maketitle
\bibliographystyle{plain}
\section{Introduction}\label{sec:intro}
This reference stays in this paper: \ref{sec:intro}.
\bibliography{second-references}
\end{document}
```

5.2. Compilation and bibliography workflows

For a single paper with one bibliography in `paper.tex`, use:

```
latex paper
bibtex paper
latex paper
latex paper
```

The class writes a bibliography unit such as `paper1.aux` and includes it from `paper.aux` when exactly one paper and one bibliography request have been encountered. No paper count is saved for a later run. BibTEX produces `paper.bbl`; the class uses that file as a fallback when the unit-specific `paper1.bbl` is absent. If both files exist, the unit-specific file takes precedence. After switching workflows, remove or regenerate a stale unit-specific `.bbl` rather than expecting the fallback to override it.

For a collection, run BibTEX on the individual bibliography units:

```
latex paper
bibtex paper1
bibtex paper2
. . .
latex paper
latex paper
```

These numbers identify bibliography units. To run BibTEX on all bibliography units using the WinEdt button, use the following workaround:

```
\documentclass[10pt,twocolumn]{ijgtp}
\iffalse%
\usepackage{bibtopic}
\fi%
```

5.3. Importing external articles

The command `\inputpaper{file.tex}`, used in the document body, executes a manuscript in a group, ignores its document class declaration, and stops at its own `\end{document}`. Several articles can be combined in a single document, together with their publication information, as follows:

```
\documentclass[10pt,twocolumn]{ijgtp}
\iffalse%
\usepackage{bibtopic}
\fi%

\usepackage{cmap,amsmath,amssymb}
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage[english]{babel}
\usepackage{epstopdf}
\epstopdfsetup{suffix=}

\begin{document}

\arxivnumber{2701.00001}
\receiveddate{11 January 2027}
\reviseddate{24 January 2027}
\accepteddate{26 January 2027}
\publisheddate{22 January 2027}
\ijgtppublished{2027}{3}{1}{1}
\inputpaper{paper1/main}

\arxivnumber{2702.00001}
\receiveddate{11 February 2027}
\reviseddate{24 February 2027}
\accepteddate{26 February 2027}
\publisheddate{22 February 2027}
\ijgtppublished{2027}{3}{1}{2}
\inputpaper{paper2/article.tex}

\arxivnumber{2703.00001}
\receiveddate{11 March 2027}
\reviseddate{24 March 2027}
\accepteddate{26 March 2027}
\publisheddate{22 March 2027}
\ijgtppublished{2027}{3}{1}{3}
\inputpaper{paper3/manuscript}

\end{document}
```

The external manuscript can be prepared using the `ijgtp`, `article`, `revtex`, or `iopjournal` class, and possibly other popular classes for scientific papers.